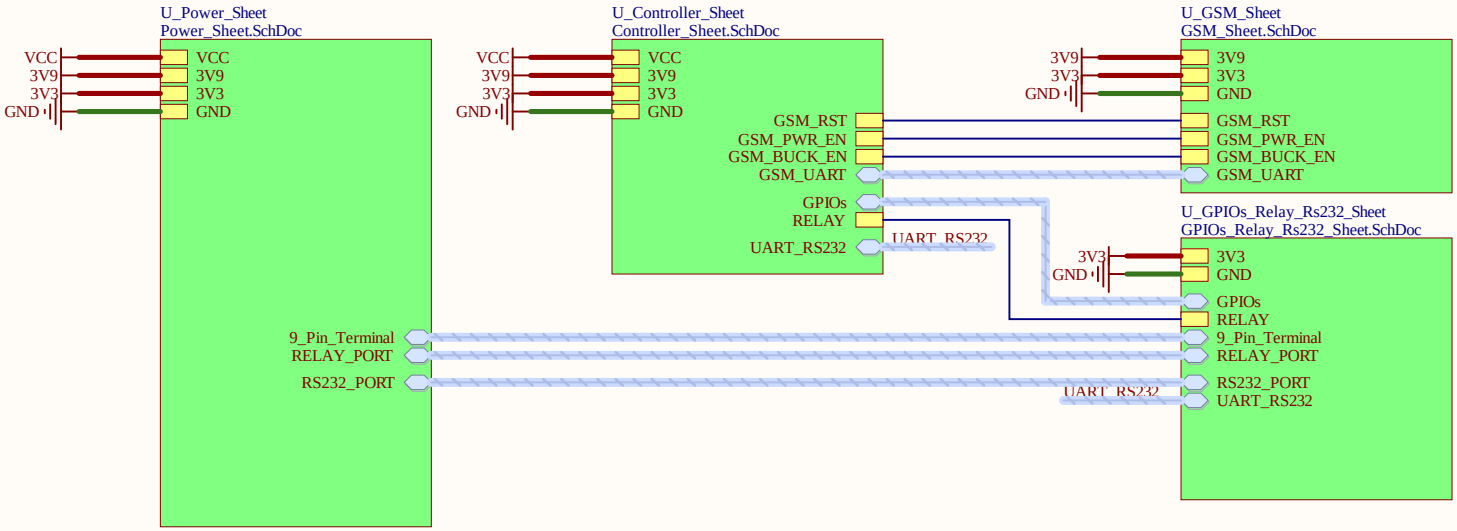
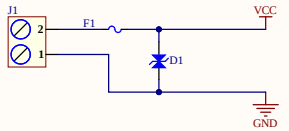


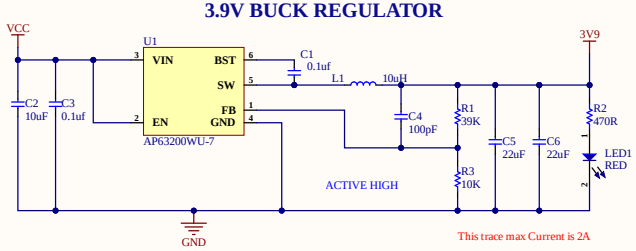
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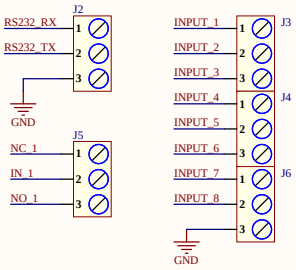
9-30V DC INPUT



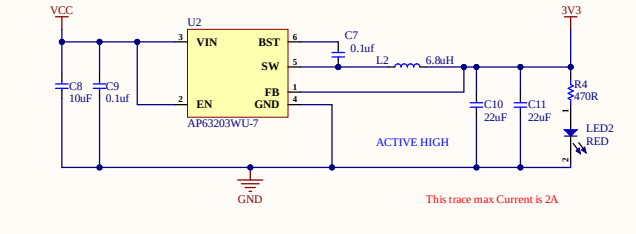
3.9V BUCK REGULATOR

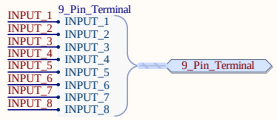


SCREW TERMINALS



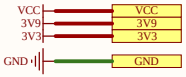
3V3 BUCK REGULATOR



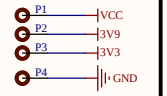






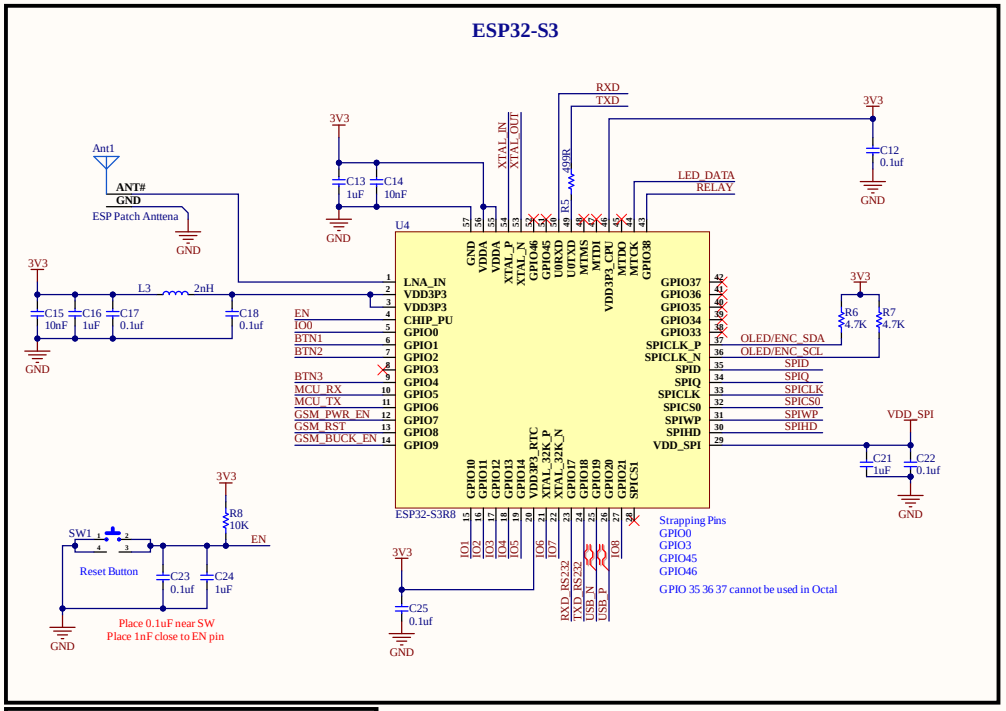


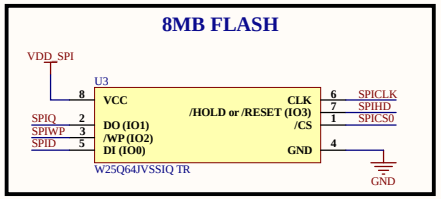
TEST POINTS

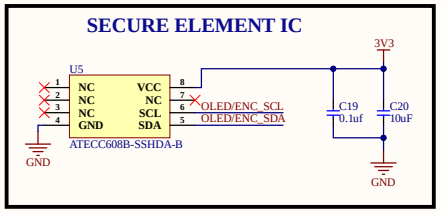


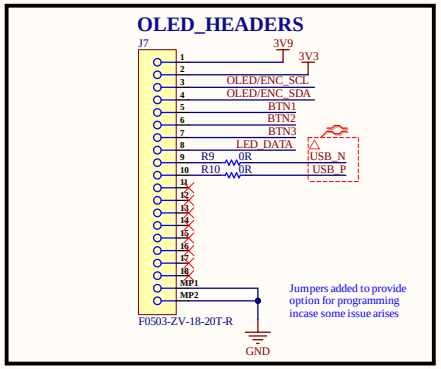
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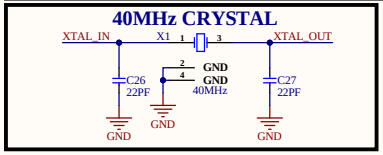
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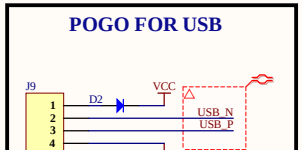


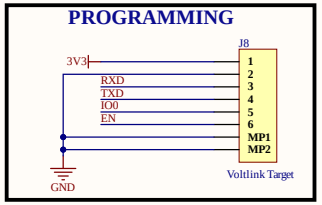


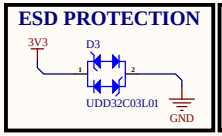


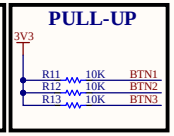


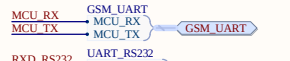








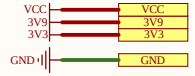








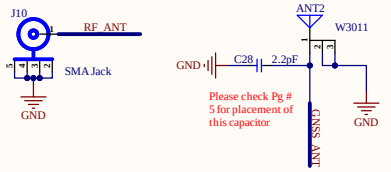




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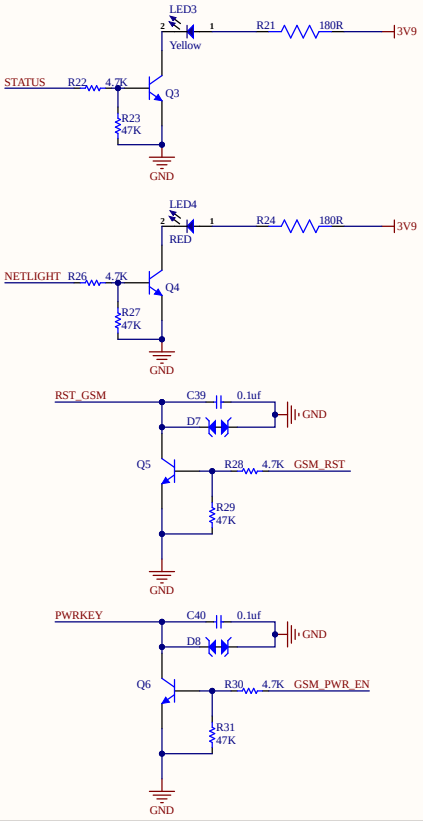
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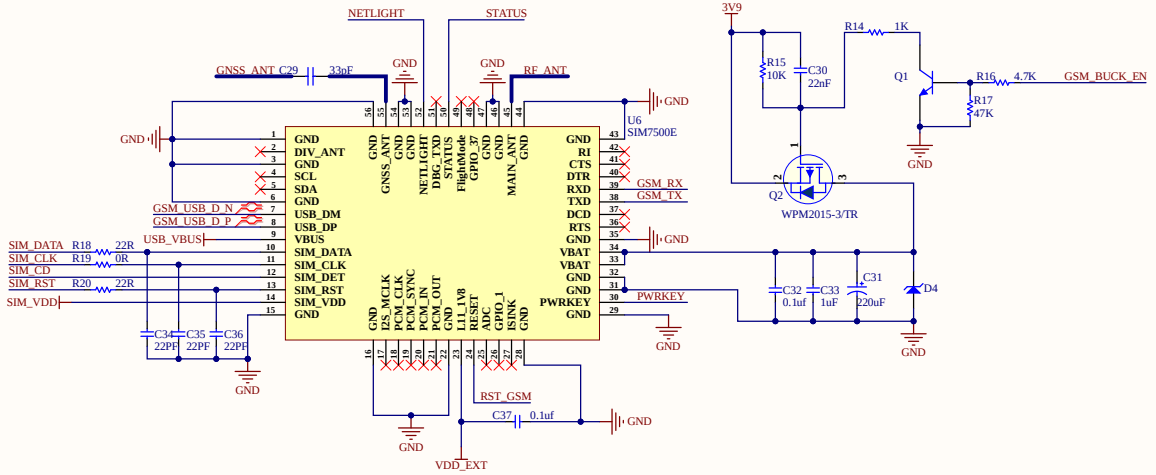


Ground cleared under antenna, clearance area 4.00 x 4.25 mm
Feed line should be designed to match 50 ohm characteristic impedance, depending on PWB material and thickness.
Recommended test board layout for electrical characteristic measurement, test board outline size 80 x 37 mm.
For more info check datasheet

GSM_COMMANDS



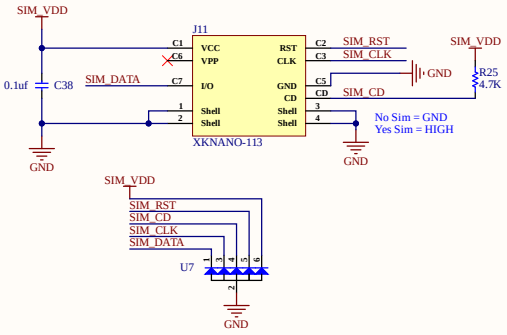
SIM7500E



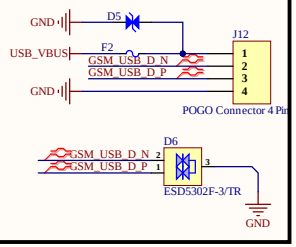
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

Datasheet Pg 31

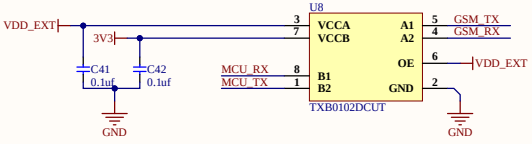
NANOSIM CONNECTOR

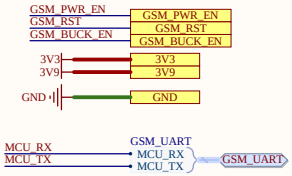


USB POGO for Updates



Level Shifter for GSM Serial

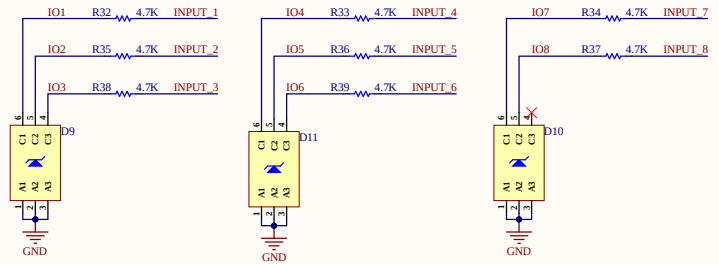




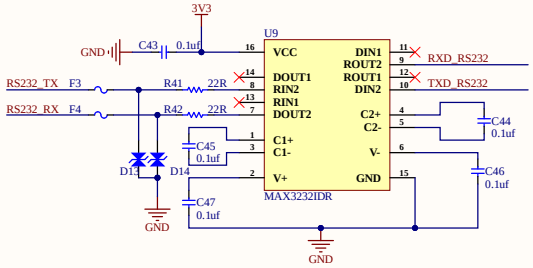
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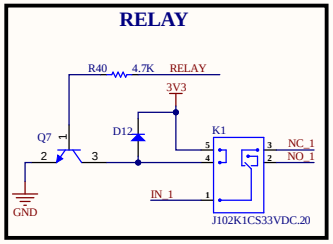
OPTO ISOLATION FOR GPIOs



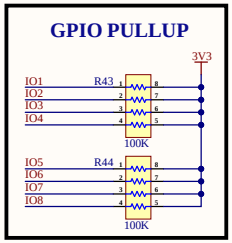
RS232 to UART

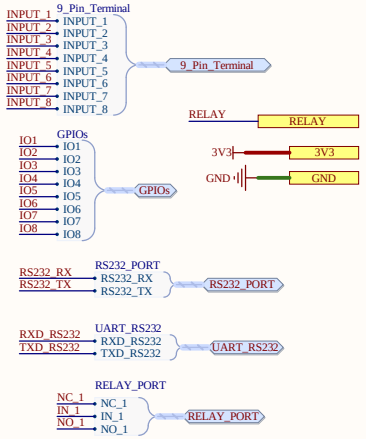


RELAY



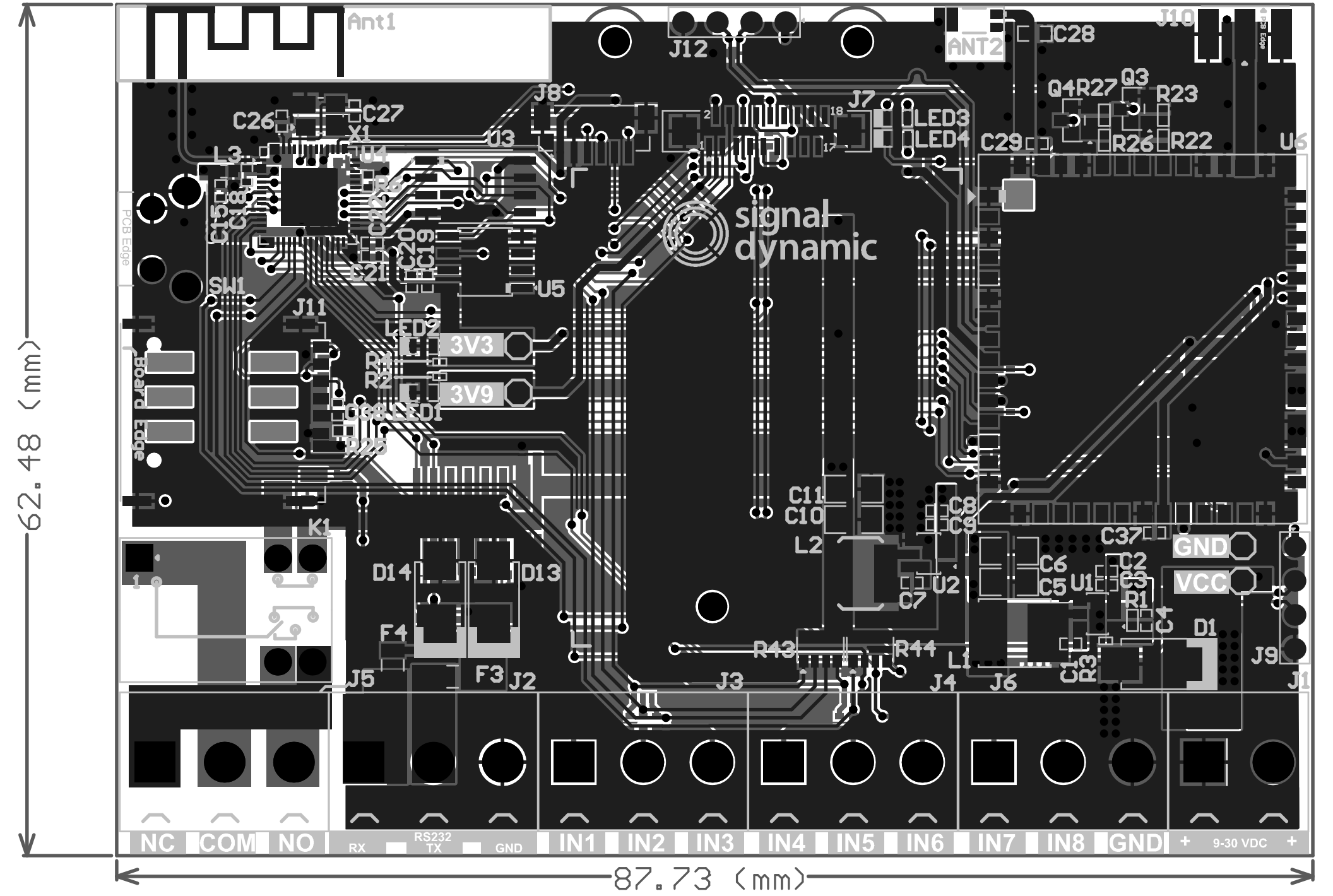
GPIO PULLUP





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Date: 8/2/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 5


62.48 (mm)
87.73 (mm)
Ant1
ANT2
J12
J10
J8
J7
C28
Q4
R27
Q3
R23
C26
C27
X1
U3
U4
L3
LED3
LED4
C29
R26
R22
U6
C15
C18
C20
C19
C21
U5
signal
dynamic
SW1
J11
LED2
3V3
3V9
R4
R2
LED1
R25
PCB Edge
Board Edge
K1
D14
D13
F4
F3
J5
J2
J3
J4
J6
C11
C10
C8
C9
L2
C37
GND
VCC
C6
C5
C7
U2
C2
C3
U1
R1
C4
D1
J9
J1
R43
R44
L1
C12
C13
NC
COM
NO
RX
RS232
TX
GND
IN1
IN2
IN3
IN4
IN5
IN6
IN7
IN8
GND
+
9-30 VDC
+